



Labor productivity, internal pay disparity, and organizational resilience

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ABSTRACT

Organizational resilience is essential for firms navigating volatile business environments. This study examines the impact of labor productivity on resilience, emphasizing the moderating role of internal pay disparity. Using a panel dataset of Chinese listed firms (2010–2022), the results show that labor productivity enhances resilience, but excessive pay disparity weakens this effect by undermining employee morale. Heterogeneity analysis reveals that firm ownership, life cycle stage, and industry classification influence this relationship. These findings emphasize the need for balanced human capital management strategies to enhance resilience, providing valuable insights for managers and policymakers to optimize workforce efficiency and pay structures.

1. Introduction

Firms' ability to withstand, adapt to, and recover from disruptions has become increasingly critical in the rapidly evolving global landscape. This capacity, which is commonly referred to as organizational resilience, enables firms to maintain stability and sustain long-term competitive advantage, even in the face of economic shocks, technological disruptions, and regulatory changes (Lengnick-Hall et al., 2011; Wu et al., 2025). As markets become more volatile due to globalization, digital transformation, and geopolitical uncertainties, firms must develop resilience through financial and operational capabilities as well as effective human capital management. Despite extensive research on resilience, the influence of labor productivity as a key determinant of an organization's adaptive capacity remains underexplored, particularly in conjunction with internal pay disparity; a structural factor that may either reinforce or undermine the productivity–resilience link.

Labor productivity is a critical lens for understanding organizational efficiency, reflecting the extent to which firms can maximize output relative to labor inputs. Rooted in the resource-based view (RBV) (Barney, 1991), a highly productive workforce represents a valuable, rare, inimitable, and nonsubstitutable (VRIN) resource that contributes to enterprises' competitive advantages and resilience (Lengnick-Hall et al., 2011). According to RBV, firms with greater labor productivity have stronger adaptive capabilities, as they can generate higher financial returns, maintain operational flexibility, and respond to environmental shifts more effectively (Crook et al., 2011). However, productivity alone does not guarantee resilience, and its effects are contingent on how firms manage, motivate, and

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distribute rewards among employees.

To extend the RBV perspective, this study incorporates dynamic capabilities theory, which suggests that firms must continuously sense, seize, and reconfigure resources to remain resilient in uncertain environments (Teece et al., 1997). A highly productive workforce enhances dynamic capabilities by fostering innovation, rapid problem-solving, and adaptive learning, which enables firms to adjust business models and operational strategies in response to disruptions (Wang and Ahmed, 2007). However, organizational resilience is not solely a function of productivity. It is also shaped by internal fairness and incentive structures as workforce cohesion and motivation influence the extent to which employees contribute to firm-wide adaptability.

In this regard, internal pay disparity, referring to the gap between executive compensation and employee wages, is a moderator that may enhance or erode resilience. Equity theory (Adams, 1963) suggests that employees assess compensation fairness relative to their peers, which directly affects their motivation, commitment, and willingness to engage in discretionary efforts (Colquitt, 2001). Excessive pay disparity can establish perceptions of unfairness, reducing employees' cooperation, lowering trust in leadership, and diminishing organizational commitment (Shaw et al., 2013). In the context of resilience, firms with high pay disparities may struggle to leverage labor productivity effectively as employee disengagement and turnover risks offset the benefits of a highly efficient workforce. Conversely, tournament theory (Lazear and Rosen, 1981) offers an alternative perspective, arguing that moderate pay disparity can incentivize performance by establishing aspirational targets for employees. From this perspective, controlled compensation inequality may motivate employees to enhance their productivity, strengthening firms' resilience (Connelly et al., 2014).

Considering these theoretical insights, we investigate the complex relationship between labor productivity and organizational resilience, emphasizing how internal pay disparity moderates this effect. By integrating RBV, dynamic capabilities, equity, and tournament theories, we provide a more holistic understanding of how firms can enhance resilience through strategic human capital management. Specifically, this study makes three key contributions: (1) empirically establishing the link between labor productivity and resilience, (2) analyzing how pay disparity influences this relationship, and (3) identifying firm-level factors that shape resilience dynamics. Using a panel dataset of Chinese listed firms from 2010 to 2022, we present robust empirical evidence to inform theoretical debates and practical strategies for enhancing workforce-driven resilience.

By addressing these research gaps, this study contributes to the ongoing discourse on organizational resilience, particularly in the domains of human capital management, labor economics, and corporate governance. In an era marked by economic uncertainty and intensifying global competition, understanding how firms can leverage labor productivity while maintaining equitable pay structures is not only a theoretical imperative but also a practical necessity for long-term survival and success.

2. Theoretical review and hypotheses development

2.1. Labor productivity and organizational resilience

Firms' RBV (Barney, 1991) provides a foundational lens for understanding the correlation between labor productivity and organizational resilience. RBV posits that firms achieve competitive advantages by possessing and effectively leveraging VRIN resources. Human capital, particularly a highly productive workforce, is a valuable and potentially rare resource that contributes to firms' competitive advantage and resilience (Lengnick-Hall et al., 2011). Highly productive employees are more likely to generate innovative solutions, adapt to changing circumstances, and contribute to the organization's overall efficiency and effectiveness, enhancing resilience (Crook et al., 2011).

Furthermore, the dynamic capabilities perspective (Teece et al., 1997) complements the RBV by emphasizing firms' ability to integrate, build, and reconfigure internal and external competencies to address rapidly evolving environments. Labor productivity has a crucial role in developing and deploying dynamic capabilities. A productive workforce is more capable of identifying, leveraging, and reconfiguring resources to maintain competitive advantages and navigate disruptions (Wang and Ahmed, 2007). For instance, a highly productive workforce can quickly adapt to new technologies, implement process improvements, and respond to market shifts, enabling organizations to remain resilient in the face of uncertainty.

Based on these theoretical perspectives, we propose the following hypothesis.

Hypothesis 1: Labor productivity is positively associated with organizational resilience.

2.2. Moderating role of internal pay disparity

While labor productivity can enhance organizational resilience, its impact may be contingent on how the firm manages human resources. Equity theory (Adams, 1963) contends that individuals are motivated by a sense of fairness in the workplace. Internal pay disparity can influence employees' perceptions of equity and affect motivation, cooperation, and organizational commitment (Colquitt, 2001). Excessive pay disparity may establish employees' feelings of unfairness and demotivation, potentially hindering organizational performance and resilience (Shaw et al., 2013).

However, tournament theory (Lazear and Rosen, 1981) proposes a contrasting perspective, suggesting that pay disparity can be a motivator by increasing competition and incentivizing employees to strive for higher positions and better performance. Moderate pay disparity may be essential for attracting and retaining talent, encouraging effort, and improving organizational effectiveness (Connelly et al., 2014). Therefore, the relationship between labor productivity and organizational resilience may be strengthened when pay disparity is moderate as it can incentivize employees to use their skills and contribute to the firm's ability to adapt and thrive. Based on these theoretical perspectives, we propose the following hypothesis.

Hypothesis 2: Internal pay disparity moderates the relationship between labor productivity and organizational resilience, and the

positive relationship is stronger when internal pay disparity is moderate.

3. Methodology

3.1. Sample selection and data source

This study's sample includes listed companies in China from 2010 to 2022. Financial data are obtained from the China Stock Market & Accounting Research database, which provides comprehensive financial information on publicly listed Chinese companies. We exclude firms with missing data on key variables from the sample to ensure data quality and reliability. The final sample consists of 27,174 firm-year observations.

3.2. Variable measurement

We examine the impact of labor productivity on organizational resilience, with internal pay disparity as a moderating variable. The study also introduces several control variables to account for potential confounding factors. The measurement of each variable is described below.

3.2.1. Explanatory variable

Labor productivity (LABORPRD): We measure this variable as the natural logarithm of firms' output per employee, which we calculate by dividing operating income by the number of employees (Tian and Estrin, 2008).

3.2.2. Explained variable

Organizational resilience (Resilience): We calculate this variable based on firms' sales revenue relative to the average sales revenue of all firms in the sample. This measure captures firms' ability to maintain performance relative to industry peers when confronting economic fluctuations (Hillmann, and Guenther, 2021). Specifically, this variable is determined as follows:

$$Resilience = ((\Delta Ave_{-}(t-1) / \Delta Ave_{-}t)) / ((\Delta TotalAve_{-}(t-1) / \Delta TotalAve_{-}t))$$

where *Ave* represents firms' sales revenue, and *TotalAve* represents the average sales revenue of all firms in the sample.

3.2.3. Alternative measure of organizational resilience (Resilience_replace)

Referencing Ortiz-de-Mandojana and Bansal (2016), this alternative measure of organizational resilience is a composite index that captures high performance growth and low financial volatility. Performance growth is measured as cumulative sales revenue growth over three years, and financial volatility is measured as the standard deviation of monthly stock returns within a year. We standardize these two indicators and then combine them using an entropy weighting method to calculate a comprehensive organizational resilience score.

3.2.4. Moderating variable

Internal pay disparity (Gap): We calculate this variable as the ratio of average management compensation to average employee compensation (Bloom and Van Reenen, 2010). Management compensation includes the total compensation of all directors and supervisors, and employee compensation is calculated as total employee compensation minus management compensation divided by the number of employees.

To isolate the effects of labor productivity and internal pay disparity on organizational resilience, we include several control variables in the model. These variables, which were previously identified as significant determinants of firm performance and resilience include firm size (Size), measured as the natural logarithm of total assets; asset-liability ratio (RDA); return on equity (ROE) (Waddock and Graves, 1997); cash flow ratio (Cashflow) (Ortiz-de-Mandojana and Bansal, 2016); financial leverage (FL) (Armstrong and Shimizu, 2007); Tobin's Q (TobinQ), which is a measure of firm value (Shin and Stulz, 1998); financing constraints (SA) (Hadlock and Pierce, 2010); and board size (Board) (Linck et al., 2008).

3.3. Model specification

We first employ the following baseline regression model to test our hypotheses.

$$Resilience_{it} = \beta_0 + \beta_1 LABORPRD_{it} + Control\ Variables_{it} + \varepsilon_{it}$$

We then construct the following moderating model using the interaction term:

$$Resilience_{it} = \gamma_0 + \gamma_1 LABORPRD_{it} + \gamma_2 Gap_{it} + \gamma_3 LABORPRD_{it} \times Gap_{it} + Control\ Variables_{it} + \varepsilon_{it}$$

where $Resilience_{it}$ represents the organizational resilience of firm i in year t . $LABORPRD_{it}$ denotes the labor productivity of firm i in year t . Gap_{it} represents the internal pay disparity of firm i in year t . $Control\ Variables_{it}$ denotes a vector of control variables for firm i in year t . ε_{it} is the error term.

We also conduct several robustness tests to ensure the robustness of our findings, using the alternative measure of organizational

resilience (Resilience_replace), introducing board size as a control variable, and employing alternative model specifications.

4. Results and discussions

4.1. Descriptive statistics

Table 1 presents the descriptive statistics for the key variables in our analysis. The average LABORPRD is 13.85, with a standard deviation of 0.814, indicating some variation across firms. The average Gap is 5.080, with a standard deviation of 3.590, revealing significant differences across firms. The Resilience variable exhibits substantial variation, with a mean of 1.287 and a standard deviation of 3.861.

4.2. Baseline regression

Table 2 presents the baseline regression results, providing robust evidence supporting Hypothesis 1. This result aligns with the core tenets of the RBV, indicating that a highly productive workforce can be considered a valuable and rare resource, contributing to firms' competitive advantage and capacity to withstand shocks (Barney, 1991; Lengnick-Hall et al., 2011). Productive employees are more likely to be efficient, innovative, and adaptable, enabling firms to respond to unexpected challenges effectively and maintain performance (Crook et al., 2011).

Furthermore, the positive relationship between labor productivity and organizational resilience can be interpreted through the lens of dynamic capabilities theory (Teece et al., 1997). Labor productivity enhances these dynamic capabilities by providing firms with the capacity to quickly adjust to new technologies, implement process improvements, and respond to market shifts (Wang and Ahmed, 2007). For instance, a highly productive workforce can quickly adopt new production methods or reskill to meet evolving customer demands, enabling the firm to remain resilient when facing disruption.

The economic implications of this finding are significant. In a competitive landscape characterized by increasing uncertainty and volatility, firms with higher labor productivity are better positioned to navigate challenges and sustain performance. This verifies the importance of investments in human capital, employee training, and work practices that promote efficiency and effectiveness. Firms can build a crucial source of resilience by fostering a highly productive workforce, enabling them to not only survive but thrive in the face of adversity.

However, it is crucial to acknowledge that labor productivity is not the sole determinant of organizational resilience. Other factors, such as leadership quality, organizational culture, and external environmental factors, also have a crucial influence. Furthermore, the relationship between labor productivity and resilience may be complex and contingent on various organizational factors, which we explore in the subsequent moderating effect analysis of internal pay disparity.

4.3. Moderating effect

Table 3 presents the results of the moderating regression that examines the impact of internal pay disparity on the relationship between labor productivity and organizational resilience. The coefficient of the Gap $\times$ LABORPRD interaction term is negative and statistically significant ($p < 0.001$) in models (1) and (2), indicating that internal pay disparity negatively moderates the relationship between labor productivity and organizational resilience. Specifically, the positive effect of labor productivity on organizational resilience is weaker when internal pay disparity is higher. This result partially supports Hypothesis 2, indicating that while moderate pay disparity may incentivize performance, excessive pay disparity can be detrimental to organizational resilience, which aligns with equity theory (Adams, 1963).

The main effect of Gap is positive and significant, indicating that higher pay disparity is associated with greater organizational resilience, which seemingly contradicts the moderating effect and may be explained by tournament theory (Lazear and Rosen, 1981). However, the negative and significant interaction effect indicates that the benefits of pay disparity in incentivizing performance may be offset by its negative impact on employee morale and cooperation when the disparity becomes excessive.

Table 1
Descriptive statistics.

Variables	N	Mean	p50	SD	Min	Max
Resilience	27,174	1.287	0.831	3.861	−10.91	30.91
LABORPRD	27,174	13.85	13.76	0.814	12.18	16.37
Gap	27,174	5.080	4.058	3.590	0.890	21.58
Size	27,174	22.26	22.05	1.279	20.11	26.07
RDA	27,174	0.406	0.401	0.194	0.0580	0.842
ROE	27,174	0.0730	0.0750	0.102	−0.403	0.343
Cashflow	27,174	0.0510	0.0490	0.0650	−0.132	0.228
FL	27,134	1.231	1.043	0.772	−0.382	5.784
TobinQ	27,174	1.987	1.610	1.161	0.844	7.301
SA	27,174	−3.808	−3.810	0.249	−4.405	−3.211

Table 2
Baseline regression.

Variables	(1) Resilience	(2) Resilience
LABORPRD	1.006*** (30.100)	0.810*** (23.256)
Size		−0.322*** (−12.861)
RDA		2.248*** (14.427)
ROE		10.840*** (43.416)
Cashflow		−2.748*** (−7.126)
FL		−0.171*** (−5.646)
TobinQ		0.222*** (9.639)
SA		0.820*** (8.157)
Constant	−12.874*** (−21.877)	−2.675*** (−3.241)
Observations	27,174	27,134
R ²	0.074	0.152
Ind	FE	FE
Year	FE	FE

Note: t-statistics are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3
Moderating effect tests.

Variables	(1) Resilience	(2) Resilience
LABORPRD	0.846*** (23.173)	0.835*** (22.810)
Gap × LABORPRD		−0.035*** (−4.574)
Gap	0.024*** (3.287)	0.024*** (3.342)
Size	−0.354*** (−13.193)	−0.349*** (−13.018)
RDA	2.229*** (14.299)	2.229*** (14.307)
ROE	10.704*** (42.296)	10.712*** (42.345)
Cashflow	−2.827*** (−7.318)	−2.894*** (−7.488)
FL	−0.164*** (−5.414)	−0.164*** (−5.423)
TobinQ	0.221*** (9.579)	0.224*** (9.718)
SA	0.822*** (8.181)	0.876*** (8.661)
Constant	−2.577*** (−3.121)	−2.423*** (−2.933)
Observations	27,134	27,134
R ²	0.153	0.153
Ind	FE	FE
Year	FE	FE

Note: t-statistics are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.4. Robustness tests

4.4.1. Endogeneity tests

Endogeneity or a two-way causal relationship between variables could bias our estimates if unobserved factors influence labor productivity and organizational resilience. We employ several strategies to mitigate these concerns.

Lagged labor productivity. We introduce a lagged labor productivity variable (L.LABORPRD) into the model (Table 4, column (1)) to reduce the potential for reverse causality. The results align with our previous findings, with the lagged variable showing a positive

and significant impact on resilience.

Instrumental variable approach. We employ the instrumental variable (IV) approach (Table 4, column (2)) using the lagged value of industry average labor productivity as an IV for current labor productivity. This IV is likely correlated with firm labor productivity but not directly with firm resilience, meeting the criterion for a valid instrument. The IV results reinforce the positive link between labor productivity and resilience.

Heckman selection model. We apply a Heckman selection model (Table 4, column (3)) to account for any potential sample selection bias. The results hold, with labor productivity continuing to exhibit a positive and significant effect on resilience.

Propensity score matching (PSM). We use PSM (Table 4, column (4)) to construct a matched sample of firms with similar characteristics but varying labor productivity levels. The results from the matched sample further validate the positive association between labor productivity and resilience.

4.4.2. Robustness tests

In addition to addressing potential endogeneity concerns, we conduct additional robustness tests to ensure that our results are not sensitive to specific model choices or variable measurements.

Alternative dependent variable. We rerun the analysis using the alternative measure of organizational resilience (Resilience_alter) (Table 5, column (1)), with a composite index that captures performance growth and financial volatility. The results using this alternative resilience measure validate our baseline findings.

Alternative model specification. We conduct an analysis using industry and year FE dummies but excluding firm FE (Table 5, column (2)), and the results remain robust.

Additional control variable. We introduce board size (Board) as an additional control variable (Table 5, column (3)) to consider potential governance effects, and this addition did not alter the baseline findings.

The consistent results across our diverse robustness tests provide strong support for the baseline findings, indicating that the positive relationship between labor productivity and organizational resilience is robust and not driven by endogeneity or model specification concerns.

4.5. Heterogeneity analysis

To deepen our examination of the relationship between labor productivity and organizational resilience, we conduct heterogeneity analyses, dividing our sample into different groups based on firm ownership and life cycle stage.

Table 4
Endogeneity tests.

Variables	(1) LABORPRD	(2) Resilience	(3) Heckman Resilience	(4) PSM Resilience
L.LABORPRD	0.916*** (212.294)			
LABORPRD		0.073* (1.810)	1.123*** (21.520)	0.805*** (15.808)
Imr1			4.006*** (8.037)	
Size	0.004* (1.871)	−0.201*** (−7.841)	−0.516*** (−14.848)	−0.235*** (−5.681)
RDA	0.132*** (9.439)	2.499*** (13.492)	3.737*** (15.444)	2.032*** (8.813)
ROE	0.471*** (19.495)	11.163*** (34.833)	22.582*** (15.238)	10.645*** (28.151)
Cashflow	0.008 (0.212)	−2.962*** (−5.848)	−6.332*** (−10.746)	−3.039*** (−5.284)
FL	0.001 (0.597)	−0.171*** (−5.382)	−0.241*** (−7.675)	−0.188*** (−4.137)
TobinQ	0.001 (0.788)	0.267*** (8.454)	0.456*** (12.288)	0.281*** (8.186)
SA	0.019** (2.416)	0.867*** (8.446)	1.641*** (11.456)	1.093*** (6.777)
Constant	1.080*** (14.218)	5.848*** (6.218)	−4.603*** (−5.362)	−3.127** (−2.378)
Observations	22,129	22,129	27,134	12,223
R ²	0.903	0.140	0.154	0.165
Ind	FE	FE	FE	FE
Year	FE	FE	FE	FE
ID			NO	NO

Note: t-statistics are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5
Additional robustness tests.

Variables	(1) Resilience_alter	(2) Resilience	(3) Resilience
LABORPRD	0.001** (2.036)	1.567*** (22.603)	0.811*** (23.244)
Size	0.004*** (20.676)	0.141* (1.958)	−0.323*** (−12.651)
RDA	−0.020*** (−16.920)	3.408*** (12.253)	2.246*** (14.415)
ROE	0.000 (0.200)	11.649*** (38.171)	10.845*** (43.422)
Cashflow	0.022*** (7.629)	−0.709 (−1.551)	−2.753*** (−7.138)
FL	−0.000 (−1.019)	−0.214*** (−6.364)	−0.171*** (−5.642)
TobinQ	−0.006*** (−34.480)	0.181*** (5.566)	0.222*** (9.629)
SA	−0.001* (−1.934)	0.088 (0.199)	0.821*** (8.151)
Board			0.027 (0.224)
Constant	0.810*** (132.393)	−25.568*** (−11.075)	−2.715*** (−3.227)
Observations	26,718	27,134	27,133
R ²	0.745	0.147	0.152
Ind	FE	NO	FE
Year	FE	FE	FE
ID	NO	FE	NO

Note: t-statistics are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

4.5.1. Firm ownership

We divide the sample into state-owned enterprises (SOEs) and non-SOEs (NSOEs) to examine whether the impact of labor productivity on resilience differs across ownership structures (Panel A, Table 6). The results reveal that while labor productivity is positively associated with resilience in both groups, the coefficient is larger for SOEs (0.902) than NSOEs (0.788). This implies that labor productivity may be a more critical driver of resilience for SOEs, which is potentially due to their unique institutional context and strategic objectives. SOEs often operate in strategic industries and face greater government and public scrutiny, making resilience crucial for legitimacy and survival.

4.5.2. Life cycle stage

We next classify firms into three life cycle stages—growth, maturity, and decline—based on sales revenue growth patterns (Panel B, Table 6). The results reveal that the impact of labor productivity on resilience varies significantly across different stages, with the strongest positive effect on resilience in the growth stage (0.949), followed by the maturity stage (0.107), and the weakest effect in the decline stage (0.112). This finding indicates that labor productivity is particularly crucial for building resilience in firms' early life cycle stages when they are more vulnerable to environmental shocks and resource constraints. As firms mature and gain market share, other factors such as organizational structure and strategic choices may become relatively more important for maintaining resilience. In the decline stage, firms may face challenges in maintaining labor productivity, and their resilience may depend more on factors such as cost-cutting measures and asset restructuring.

4.5.3. Industry characteristics

Industry characteristics have an influence on the impact of labor productivity on organizational resilience (Panel C, Table 6). Firms in nonheavy polluting industries exhibit a strong and positive relationship between labor productivity and resilience ($\beta = 0.507$, $p < 0.01$), indicating that workforce efficiency directly translates into adaptability and sustained performance. With fewer regulatory constraints, these firms can leverage labor productivity to drive innovation, maintain stability, and enhance operational flexibility. Employees in such industries may also benefit from greater job security and engagement, reinforcing firms' ability to navigate disruptions.

In contrast, firms in heavy-polluting industries experience a significantly weaker positive effect ($\beta = 0.143$, $p < 0.05$), indicating that while labor productivity remains important, its benefits are constrained by strict environmental regulations, sustainability mandates, and industry volatility. High compliance costs and operational uncertainties can limit managerial flexibility, making it harder to fully capitalize on labor efficiency. Additionally, workforce morale may be affected by job insecurity and regulatory pressure. To enhance resilience, firms in these sectors must integrate sustainability-driven strategies, such as environmental innovation and corporate social responsibility initiatives, with effective human capital management.

The heterogeneity analysis provides insights into how the relationship between labor productivity and organizational resilience varies across different firm characteristics of ownership structure, life cycle stage, and industry classification. First, the results

Table 6
Heterogeneity tests.

PANEL A		Firm ownership	
Groups		NSOE	SOE
Variables		Resilience	Resilience
LABORPRD		0.788*** (17.097)	0.902*** (16.368)
Size		−0.194*** (−5.497)	−0.341*** (−8.043)
RDA		2.570*** (12.889)	2.010*** (7.628)
ROE		11.343*** (37.064)	9.565*** (20.725)
Cashflow		−3.598*** (−7.451)	−1.566** (−2.376)
FL		−0.251*** (−5.789)	−0.070* (−1.655)
TobinQ		0.216*** (7.908)	0.133*** (2.888)
SA		1.002*** (7.447)	0.332* (1.900)
Constant		−4.843*** (−4.473)	−4.504*** (−3.018)
Observations		17,548	9040
R ²		0.179	0.127
Ind		FE	FE
Year		FE	FE
PANEL B Life cycle stage			
Groups	Growth stage	Maturity stage	Declining stage
Variables	Resilience	Resilience	Resilience
LABORPRD	0.949*** (16.183)	0.107** (2.510)	0.112* (1.904)
Size	−0.375*** (−8.947)	−0.126*** (−4.408)	−0.081* (−1.727)
RDA	1.867*** (6.580)	1.546*** (8.582)	1.712*** (6.504)
ROE	11.572*** (26.509)	8.738*** (28.229)	9.264*** (24.150)
Cashflow	−2.333*** (−3.265)	−0.448 (−0.806)	−1.676*** (−2.596)
FL	−0.229*** (−4.732)	−0.052 (−1.343)	−0.194*** (−3.897)
TobinQ	0.344*** (8.277)	0.214*** (8.539)	0.084** (2.063)
SA	0.897*** (5.217)	0.477*** (4.218)	0.881*** (4.984)
Constant	−3.207** (−2.286)	1.002 (1.066)	2.883** (2.009)
Observations	11,059	10,594	5008
R ²	0.206	0.141	0.173
Ind	FE	FE	FE
Year	FE	FE	FE
PANEL C Heavy polluted industry			
Groups	Nonheavy polluted	Heavy polluted	
Variables	Resilience	Resilience	
LABORPRD	0.507*** (13.644)	0.143** (2.255)	
Size	−0.250*** (−9.410)	−0.115*** (−2.632)	
RDA	2.389*** (14.426)	1.687*** (6.240)	
ROE	10.469*** (39.769)	11.084*** (24.148)	
Cashflow	−3.615*** (−8.830)	−4.413*** (−6.483)	
FL	−0.209*** (−5.905)	−0.074* (−1.773)	
TobinQ	0.194*** (8.220)	0.339*** (7.227)	
SA	0.755*** (7.144)	0.884*** (4.959)	

(continued on next page)

Table 6 (continued)

PANEL A	Firm ownership	
Constant	−0.533 (−0.628)	1.507 (1.088)
Observations	20,907	5797
R ²	0.164	0.150
Ind	FE	FE
Year	FE	FE

Note: t-statistics are in parentheses; *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

comparing SOEs and NSOEs indicate that labor productivity has a stronger effect on SOEs' ($\beta = 0.902, p < 0.01$) than NSOEs' resilience ($\beta = 0.788, p < 0.01$). This implies that SOEs, which often operate in strategic sectors with state support, may benefit more from productivity-driven resilience, as they have superior resource stability and long-term policy backing. Second, we find the impact of labor productivity on resilience to be life cycle stage-dependent, with the strongest effect observed in the growth stage ($\beta = 0.949, p < 0.01$), followed by the maturity stage ($\beta = 0.107, p < 0.05$), and the weakest effect in the decline stage ($\beta = 0.112, p < 0.1$). This indicates that younger firms that typically face higher market uncertainty and resource constraints rely on labor productivity more heavily to build resilience and sustain competitive advantage. Meanwhile, mature firms may depend on structural stability and strategic adaptability, whereas firms in decline may struggle to maintain productivity's positive effects amid external pressure. Finally, industry classification is found to influence the labor productivity–resilience relationship, with firms in nonheavy polluting industries exhibiting a stronger resilience effect ($\beta = 0.507, p < 0.01$) than those in heavy-polluting industries ($\beta = 0.143, p < 0.05$). The weaker effect in pollution-intensive industries implies that regulatory constraints, sustainability compliance, and operational uncertainties may limit the extent to which productivity can drive resilience. Collectively, these findings confirm the context-specific nature of labor productivity's influence on resilience, emphasizing the need for strategically tailored human capital strategies that include firm ownership structure, life cycle stage, and industry environment.

5. Conclusions and future research agenda

5.1. Conclusions

This study investigates the impact of labor productivity on organizational resilience, with a particular focus on the moderating effect of internal pay disparity. Our findings using a panel dataset of Chinese listed firms (2010–2022) confirm that labor productivity significantly enhances resilience, as a more efficient workforce enables firms to adapt to disruptions and sustain competitive performance. The results also reveal that excessive internal pay disparity weakens this relationship, indicating that disproportionate compensation gaps can undermine organizational cohesion, employee engagement, and resilience. These findings contribute to human capital management literature by emphasizing the dual role of labor productivity and pay structures in shaping organizational adaptability.

A deeper theoretical reflection reveals the interplay between RBV (Barney, 1991), dynamic capabilities theory (Teece et al., 1997), and equity theory (Adams, 1963) for explaining how firms can build resilience. From the RBV perspective, a highly productive workforce is a valuable and rare resource that strengthens competitive advantage; however, pay inequity can erode this strategic asset by diminishing employees' perceived fairness and motivation. Dynamic capabilities theory further supports our findings, confirming that firms with higher labor productivity are better equipped to leverage resources in response to external shocks. However, excessive internal pay disparity may disrupt collaboration and innovation, weakening firms' resilience. Additionally, the findings align with equity theory, which suggests that employees' perceived fairness shapes commitment, and firms that fail to maintain an optimal balance in compensation risk fostering dissatisfaction and reducing workforce resilience.

Our heterogeneity analysis demonstrates that the labor productivity–resilience relationship is context-dependent, influenced by firm ownership, life cycle stage, and industry classification. Labor productivity has a stronger effect in SOEs than NSOEs, which is likely attributable to institutional support and long-term policy stability. Moreover, firms in the growth stage benefit the most from productivity-driven resilience as they rely on workforce efficiency to navigate market uncertainties. In contrast, firms in heavy-polluting industries experience weaker effects due to regulatory constraints and compliance costs that limit their ability to fully leverage labor productivity for resilience. These insights emphasize that resilience-building strategies should be tailored to firm- and industry-specific conditions, rather than using universal approaches.

5.2. Research limitations and future directions

Despite its contributions, this study has several limitations that offer avenues for future research. First, our organizational resilience measurement is based on financial performance indicators, which may not fully capture long-term adaptive capacity, cultural resilience, or leadership-driven responses. Future studies could use qualitative or multidimensional resilience measures such as employee retention rates, crisis response mechanisms, and innovation outputs. Second, our dataset focuses exclusively on Chinese listed firms, limiting generalizability to other institutional contexts. Considering that labor market structures, regulatory frameworks, and corporate governance systems vary widely across countries, comparative studies of different economies would provide deeper insights.

Third, while our study explores pay disparity as a key moderator, resilience is likely influenced by multiple organizational factors such as leadership styles, corporate culture, and external economic conditions. Future research could incorporate such dimensions to develop a more comprehensive resilience framework. Finally, we assume a linear relationship between labor productivity and resilience; however, nonlinear or threshold effects may exist, particularly regarding the optimal level of pay disparity to balance incentives with fairness. Exploring such dynamics using advanced econometric models or experimental designs could refine our understanding of how human capital policies contribute to resilience in different organizational settings.

From a managerial perspective, firms seeking to enhance resilience should prioritize investments in labor productivity such as employee training, technological innovation, and process optimization. However, these efforts must be complemented by equitable pay structures to maintain workforce motivation and cohesion. Excessive pay gaps may reduce collaboration, increase turnover, and weaken resilience, offsetting the benefits of higher productivity. Therefore, managers should adopt evidence-based compensation policies to ensure that incentives drive performance without compromising fairness.

It is crucial for policymakers to develop regulatory frameworks that balance sustainability mandates with workforce resilience, particularly for heavy-polluting industries. Stricter environmental policies may inadvertently increase firms' financial pressure, limiting their ability to invest in human capital development. Developing policies that encourage labor productivity growth and environmental compliance such as green technology subsidies, workforce retraining programs, and sustainable business incentives can help firms navigate the challenges of productivity enhancement and regulatory adaptation (Xu et al., 2024; Dai, 2025).

By integrating insights regarding human capital management, organizational resilience, and labor economics, this study provides a nuanced understanding of how firms can navigate uncertainty through workforce efficiency. The findings reinforce the importance of balancing productivity and equity in shaping long-term resilience, with strategic implications for managers, policymakers, and scholars. As businesses continue to face economic volatility, technological disruptions, and regulatory changes, developing a resilient workforce through effective human capital management will remain a critical determinant of organizational survival and success.

CRediT authorship contribution statement

Yubing Zhao: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Zhe Yuan:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Junting Jia:** Validation, Methodology, Investigation, Formal analysis, Data curation.

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Data availability

Data will be made available on request.

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